Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 20, 39, 10, 12, 26.

Final Answer: 29. Range $= \max - \min = 39 - 10 = 29$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 38, 47, 25, 57, 46.

Final Answer: 32. Range $= \max - \min = 57 - 25 = 32$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 91, 86, 36, 84, 15, 44, 11.

Final Answer: 80. Range $= \max - \min = 91 - 11 = 80$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 97, 12, 10, 30, 50, 81, 49.

Final Answer: 87. Range $= \max - \min = 97 - 10 = 87$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 87, 89, 26, 14, 61, 66, 77, 99, 38.

Final Answer: 85. Range $= \max - \min = 99 - 14 = 85$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 41, 97, 11, 85, 63, 39, 69.

Final Answer: 86. Range $= \max - \min = 97 - 11 = 86$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 8, 23, 19, 12, 3, 11.

Final Answer: 20. Range $= \max - \min = 23 - 3 = 20$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 25, 9, 11, 28, 18.

Final Answer: 19. Range $= \max - \min = 28 - 9 = 19$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 16, 5, 15, 21, 20, 15.

Final Answer: 16. Range $= \max - \min = 21 - 5 = 16$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 8, 27, 20, 25, 20.

Final Answer: 19. Range $= \max - \min = 27 - 8 = 19$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 15, 20, 16, 53, 9.

Final Answer: 44. Range $= \max - \min = 53 - 9 = 44$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 58, 31, 1, 33, 54.

Final Answer: 57. Range $= \max - \min = 58 - 1 = 57$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 20, 42, 86, 77, 5, 81, 76, 99, 72.

Final Answer: 94. Range $= \max - \min = 99 - 5 = 94$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Find the range of the data set: 66, 79, 70, 57, 8, 48, 40, 2, 14.

Final Answer: 77. Range $= \max - \min = 79 - 2 = 77$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers
Find the median of the data set: 32, 10, 29, 13, 37, 36, 34, 38.

Final Answer: 33. Sorted: 10, 13, 29, 32, 34, 36, 37, 38. The mean of the two middle values $\frac{32+34}{2} = 33$.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the median of the data set: 24, 11, 20, 9, 34, 15, 43, 29.

Final Answer: 22. Sorted: 9, 11, 15, 20, 24, 29, 34, 43. The mean of the two middle values $\frac{20+24}{2} = 22$.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the median of the data set: 11, 38, 64, 14, 28, 71, 8, 43.

Final Answer: 33. Sorted: 8, 11, 14, 28, 38, 43, 64, 71. The mean of the two middle values $\frac{28+38}{2} = 33$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

Find the median of the data set: 16, 6, 55, 25, 71, 72, 77, 74.

Final Answer: 63. Sorted: 6, 16, 25, 55, 71, 72, 74, 77. The mean of the two middle values $\frac{55+71}{2} = 63$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{27}{40}$. Independent events multiply: $P(A) \times P(B) = \frac{3}{4} \times \frac{9}{10} = \frac{27}{40}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

Two independent events A and B have $P(A) = \frac{1}{5}$ and $P(B) = \frac{3}{10}$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{3}{50}$. Independent events multiply: $P(A) \times P(B) = \frac{1}{5} \times \frac{3}{10} = \frac{3}{50}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94 ✓

d. 96

Required total for 5 exams: $5 \times 88 = 440$. Current sum: $82 + 88 + 91 + 85 = 346$. Fifth score required: $440 - 346 = 94$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12 ✓

b. 13

c. 14

d. 15

Median of 7 sorted values is the 4th value: x. Mean $= \frac{4+7+9+x+14+18+20}{7} = \frac{72+x}{7}$. Equating: $7x = 72 + x \Rightarrow 6x = 72 \Rightarrow x = 12$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?

a. 84

b. 85 ✓

c. 86

d. 87

Weighted grade $= 0.20(90) + 0.30(80) + 0.50(86) = 18 + 24 + 43 = 85$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11 ✓

c. 12

d. 14

The median is 12. The lower half is $\{3, 7, 8\}$ with $Q_1 = 7$. The upper half is $\{15, 18, 21\}$ with $Q_3 = 18$. Thus IQR $= Q_3 - Q_1 = 18 - 7 = 11$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. **It is multiplied by 3** ✓

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

Multiplying by a constant c scales standard deviation by $|c| = 3$. Adding a constant shifts the center but does not change the spread. Hence, standard deviation is multiplied by 3.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$ ✓

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

A standard deviation of 0 occurs if and only if all values in the data set are identical.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$ ✓

c. $\frac{1}{4}$

d. $\frac{3}{4}$

Total marbles $= 6 + 8 + 10 = 24$. Non-blue marbles $= 6 + 10 = 16$. Probability $= \frac{16}{24} = \frac{2}{3}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?

a. $\frac{1}{6}$ ✓

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

Outcomes with sum > 9 are sums of $10, 11, 12$: $(4,6), (5,5), (6,4), (5,6), (6,5), (6,6)$ = 6 outcomes. Probability $= \frac{6}{36} = \frac{1}{6}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70 ✓

b. 0.80

c. 0.90

d. 0.20

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126 ✓

d. 3024

Selection order does not matter: $\binom{9}{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$ ✓

b. $5,040$

c. $9,000$

d. $3,024$

Choices for 1st digit (1-9) $= 9$. Choices for 2nd digit (0-9 excluding 1st) $= 9$. Choices for 3rd digit $= 8$. Choices for 4th digit $= 7$. Total $= 9 \times 9 \times 8 \times 7 = 4,536$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

b. 36

c. 72 ✓

d. 144

By the fundamental counting principle: $4 \times 6 \times 3 = 72$ combinations.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{3}{50}$

b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{3}{20}$ ✓

Independent events multiply: $P(A) \times P(B) = \frac{3}{4} \times \frac{1}{5} = \frac{3}{20}$.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{1}{6}$ and $P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{1}{20}$

d. $\frac{1}{12}$ ✓

Independent events multiply: $P(A) \times P(B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$.

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$ **b.** $\frac{4}{5}$

c. $\frac{5}{8}$ **d.** $\frac{3}{4}$ ✓

Independent events multiply: $P(A) \times P(B) = \frac{5}{6} \times \frac{9}{10} = \frac{3}{4}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{2}{3}$ and $P(B) = \frac{3}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$ **b.** $\frac{2}{5}$ ✓

c. $\frac{1}{4}$ **d.** 0

Independent events multiply: $P(A) \times P(B) = \frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two independent events A and B have $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. Find $P(A \text{ and } B)$.

a. $\frac{2}{3}$ **b.** $\frac{1}{6}$ ✓

c. 0 **d.** $\frac{3}{4}$

Independent events multiply: $P(A) \times P(B) = \frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{2}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{1}{2}$ **b.** $\frac{1}{8}$

c. $\frac{3}{10}$ **d.** $\frac{1}{3}$ ✓

Independent events multiply: $P(A) \times P(B) = \frac{5}{6} \times \frac{2}{5} = \frac{1}{3}$.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Two independent events A and B have $P(A) = \frac{5}{6}$ and $P(B) = \frac{1}{5}$. Find $P(A \text{ and } B)$.

a. $\frac{2}{5}$

b. $\frac{1}{6}$ ✓

c. $\frac{16}{25}$

d. 1

Independent events multiply: $P(A) \times P(B) = \frac{5}{6} \times \frac{1}{5} = \frac{1}{6}$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

Two independent events A and B have $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{4}$. Find $P(A \text{ and } B)$.

a. 0

b. $\frac{1}{8}$ ✓

c. $\frac{1}{3}$

d. 1

Independent events multiply: $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Find the median of the data set: 35, 22, 1, 6, 3, 49.

a. 35

b. 8

c. 29

d. **14** ✓

Sorted: 1, 3, 6, 22, 35, 49. The mean of the two middle values $\frac{6+22}{2} = 14$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A data set of 9 values has a mean of 54. A new value of 224 is added. What is the new mean?

a. 43

b. **71** ✓

c. 47

d. 138

Old total $= 9 \times 54 = 486$. New total $= 486 + 224 = 710$. New mean $= 710/10 = 71$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A data set of 9 values has a mean of 50. A new value of 340 is added. What is the new mean?

a. 140

b. **79** ✓

c. 4

d. 164

Old total $= 9 \times 50 = 450$. New total $= 450 + 340 = 790$. New mean $= 790/10 = 79$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A data set of 9 values has a mean of 29. A new value of 179 is added. What is the new mean?

a. 69

b. **44** ✓

c. 21

d. 39

Old total $= 9 \times 29 = 261$. New total $= 261 + 179 = 440$. New mean $= 440/10 = 44$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A data set of 7 values has a mean of 33. A new value of 113 is added. What is the new mean?

a. 82

b. **43** ✓

c. 97

d. 39

Old total $= 7 \times 33 = 231$. New total $= 231 + 113 = 344$. New mean $= 344/8 = 43$.

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